TEMPORAL HIDDEN-STATE MONITORING FOR EARLY DETECTION OF ANOMALOUS LLM GENERATION

A Detection-Only Framework Using Frozen GPT-2-medium

A PROJECT REPORT

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Under the guidance of
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Submitted to



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BONAFIDE CERTIFICATE

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ABSTRACT

LLMs can generate coherent and contextually relevant text, but they can generate incorrect, misleading or anomalous responses. If such behaviour occurs only after the full response, then it will be very difficult to give an early warning. We suggest the idea of studying whether the hidden-state dynamics of LLMs can be used as an early sign of the presence of anomalous generation through a detection-only approach, NeuroTrace. The framework is based on the frozen GPT-2-medium model, and is based on the monitoring of hidden representations of layers 3 to 6 in the model, with the parameters of this layer remaining unchanged during the generation, one token at a time.

NeuroTrace represents time related features: hidden-state norms, transitions between successive hidden-state vectors, cosine similarity, and relative state changes, for each token produced. These features, when combined with uncertainty measures like normalized entropy, token probability, and entropy. The features are aggregated and then fed into a simple Logistic Regression detector to compute a risk score and classify the generation as NORMAL or SUSPICIOUS. These signals are monitored at various levels of the generated response: 25%, 50%, 75% and 100%.

For the experimental study 100 questions of HaluEval were chosen to produce 200 balanced controlled condition samples. The 75% and 100% checkpoints had mean ROC AUCs of 0.7090 and 0.7030, respectively, and the trajectory-only features measured a mean ROC AUC at 25% checkpoint of 0.9095. The controlled generation setup had around 23 tokens left at the 25% time mark, suggesting early warning prior to the end of the response. These results are rather the outcome of a controlled condition-discrimination experiment, however, and not a fully validated factual hallucination detector. The study demonstrates the potential of the temporal hidden-state monitoring approach and suggests it needs to be tested further using validated answers and more diverse test sets as well as various LLM architectures.

Keywords: Large Language Models (LLMs), Hidden States, Temporal Trajectory, Anomaly Detection, Hallucination Detection, GPT-2-medium, Early Warning, Uncertainty, Logistic Regression .

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LIST OF SYMBOLS AND ABBREVIATIONS

AI	-	Artificial Intelligence
CI	-	Confidence Interval
CUDA	-	Compute Unified Device Architecture
GPT	-	Generative Pre-trained Transformer
GPU	-	Graphics Processing Unit
LLM	-	Large Language Model
ROC-AUC	-	Area Under the Receiver Operating Characteristic Curve
SD	-	Standard Deviation
$h_t^{(L)}$	-	Hidden-state vector at token t and layer L
d	-	Dimension of the hidden state
$\Delta_t^{(L)}$	-	Hidden-state transition between consecutive tokens
ϵ	-	Small number to avoid division by zero
H	-	Entropy of the token probability distribution
p_i	-	Probability of the i-th token

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

One of the most important applications of AI is the use of Large Language Models (LLMs), which can understand instructions and generate text that is similar to that of a human. GPT-based models can generate coherent and contextually appropriate responses but the content is not necessarily accurate or trustworthy. LLMs can also produce inaccurate, misleading or out-of-context information, and provide confidence and a natural writing style. Not seeing this behavior at the final response may reduce the chance of being able to determine if there is an issue during the response creation process.

When generating, an LLM passes each token through a series of layers in a neural network, which generates hidden-state representations that capture the internal information the LLM is processing in order to generate the next token. These representations change over the course of time, generating new tokens along the way, and thus, forming a temporal sequence of internal states. NeuroTrace is exploring the possibility of using these temporal changes to give a useful early-warning signal for abnormal generation. The framework relies on a frozen GPT-2-medium model and monitors the hidden representations of layers 3 to 6, which yield different features such as hidden-state transitions, cosine similarity, relative state change, token probability and entropy. The features are analysed without altering or fine-tuning of the original language model, using a lightweight detection model to generate a risk score at various stages of generation

1.2 LARGE LANGUAGE MODELS AND ANOMALOUS GENERATION

Large Language Models are trained on large text collections, and learn patterns that allow them to generate text one token at a time. Through this process they are able to generate responses that are very coherent, fluent and relevant to the task involved for various types of tasks. The content generated might be incorrect or even contradictory and surprising, although the response sounds convincing and coherent.

An anomalous behaviour can be considered a behaviour that does not match that of the required behaviour for the required task. These cases are difficult to identify since the wrong answer might not have any linguistic errors. These are why traditional techniques that only consider the final response may only provide information at the end of the generation process, and may not afford the chance to see the model's behavior at a stage before the end of the generation process.

1.3 MOTIVATION

The need for precise monitoring of the generated content from the LLM so that the content cannot be misleading or incorrect, but sounds natural and believable is growing with the increasing use of these applications. This stimulates the learning of the model behaviour, rather than the answer itself. NeuroTrace is designed to track the dynamics of the hidden-state representations of a frozen GPT-2-medium model, and determine whether meaningful patterns appear as evidence of possible abnormal generation in early stages. The project also explores the trade-offs between precision and the time at which it is possible to receive a warning without changing the underlying language model, at 25%, 50%, 75% of the generation process.

1.4 PROBLEM DEFINITION

Even though the generated text looks fluent and convincing, LLM can produce responses that are incorrect, misleading or unexpected. When the entire response is produced, it might be too late to detect early. It is therefore important to investigate whether information about the internal behaviour of the model during its generation can be very valuable before the result of the response is completed.

NeuroTrace overcomes this issue by observing the temporal evolution of hidden-state representations of a frozen GPT-2-medium model token-by-token over time. Features such as state transitions, cosine similarity, relative state changes, token probability and entropy are used to analyse the hidden states within the layers 3-6. A light weight detection model is used to generate a risk score at various generation stages, using these features. The aim here is to investigate whether such temporal internal-states patterns can serve as an early-warning without fine-tuning, modifying or steering the language model itself.

1.5 NEED FOR INTERNAL STATE MONITORING

The difficulty with being able to observe just the answer would be that Large Language Models could give coherent, convincing answers to a problem even if it's false or aberrant. Internal state monitoring uses the hidden-state representations during model generation and in the case of NeuroTrace these representations are monitored between successive tokens to investigate if the evolution of the representations provides any advance warning.

1.6 ROLE OF HIDDEN STATES AND TEMPORAL REPRESENTATIONS IN LLMS

A hidden state is a number that is produced by the language model at each input and each output step. These representations change in the course of the generation process through evolution of the internal representation of the context that is predicted by the model. These changes can be seen from one token to the next, and these changes can be captured in the model as a temporal trajectory. During NeuroTrace, the hidden states of layers 3 –6 are monitored to investigate the dynamics of these representations throughout the generation of responses.

A temporal representation is more informative than just a single hidden state, because it depicts the difference between the generations. The distances between successive hidden-state vectors, cosine similarity, relative state change and layer-wise statistics are measured in NeuroTrace. Changes in model internal path are investigated with regard to the temporal features and the probability and entropy of the tokens to see whether any of them are correlated with the controlled conditions used for the experiment. It is not meant to imply that all changes are anomalies, but to determine whether temporal hidden-state patterns will be useful tools to use for monitoring early.

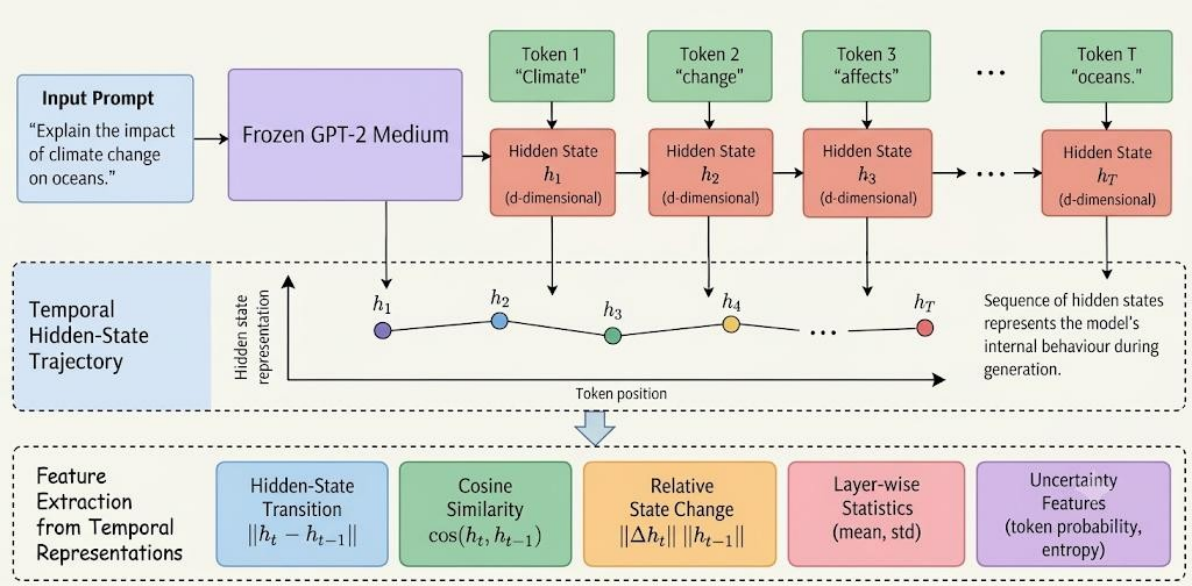


Figure 1.1: Temporal Hidden-State Representation

1.7 PROJECT OBJECTIVES

NeuroTrace aims to investigate the potential of obtaining meaningful signals, as a function of time, from internal hidden-state representations of a Large Language Model (LLM), that can be used for the early discovery of potentially anomalous generation.

It targets a frozen GPT-2-medium model, and explores hidden states of layers 3–6 when generating new tokens. The goals are to:

1. Obtain temporal features including hidden-state change, cosine similarity, relative state change and layer-wise statistics.
2. Introduce uncertainty related characteristics like probability of tokens, entropy and normalised entropy.
3. Build the light-weight logistic regression detector to calculate a risk score and classification as NORMAL/SUSPICIOUS.
4. Test the detector at 25%, 50%, 75%, and 100% generation points.

5. Calculate a time window a potential detection might occur before full response occurs.
6. Analyze trajectories based features, uncertainty based and combined features.
7. Minimize question level data leakage by testing the approach with the group aware cross validation.
8. Keep the base GPT-2-medium model completely frozen (unfined, unstabilized and unaltered).
9. Investigate the potential of internal temporal representations as early-warning

1.8 SCOPE

The focus of NeuroTrace is to explore internal hidden-state representations as a tool to monitor potentially anomalous generation of LLM. We use a "frozen" GPT-2-medium model for the study, and monitor the hidden states generated token-by-token at layers 3-6. In addition to uncertainty features (token probability, entropy), features in the form of temporal features (hidden-state transition, cosine similarity, relative state transition, layer-wise statistics) are also weighted. The features are judged at various generation points to consider if useful warning signals can be derived prior to the response.

The project is done in a research prototype manner and does not aim at changing the language model itself but at detecting language models. A light-weight Logistic Regression detector creates risk scores and observes conditions are assigned to one of two classes: NORMAL or SUSPICIOUS. The evaluation is composed of the following: Group aware Cross Validation-trajectory based, uncertainty based, combined feature sets; Early-warning and detection lead-time analysis. The scope does not address model fine tuning, activation steering, response correction or model deployment to production; the risk scores are not deemed to be confirmed factual hallucinations nor calibrated model error probabilities.

1.9 CONTRIBUTIONS

The main achievements of the NeuroTrace project are:

- Temporal hidden-state monitoring: Computation of the evolution of the hidden state representations, over time and across tokens.
- Analyses hidden states of the 3 to 6th layer of the frozen GPT-2-medium model (layer-wise).
- Hidden-state transition: Hidden-state transition of the model.
- Uncertainty analysis: Extends the features of the generation level to include the additional token probability, entropy and normalized entropy.

- Lightweight detection: Based on Logistic Regression, it gives a risk score and classifies the observed condition into NORMAL or SUSPICIOUS.
- Early-warning evaluation: Used to identify the detector at 25%, 50%, 75% and 100% of the generation process to study the early detection process.
- Lead-time analysis: This is a technique that gives a indication of the number of tokens remaining after the first warning and therefore the potential detection time.
- Feature comparison: Compares trajectory-only, uncertainty-only and combined features.
- Leakage-aware evaluation: Cross validation is done group-aware: No same question is in the training set and the test set.
- Detection-only design: Keeps the underlying model (GPT-2-medium) unfine-tuned, un-activated and without response correction.

1.10 ORGANIZATION OF THE REPORT

This report is divided into 6 chapters which present the development and evaluation of NeuroTrace in a systematic way. The project background, motivation, problem definition, need to monitor the internal state, project-objectives and scope/role/ and contributions are introduced in chapter 1. In Chapter 2 the literature is presented from which the existing research in the field of hallucination detection, internal representations, safety monitoring and early detection approaches are reviewed and a research gap is identified by NeuroTrace . This work concentrates on the problem statement and methodology, which are detailed in Chapter 3, including the datasets, controlled experimental conditions, extraction of the hidden-state features, uncertainty features, detection model, cross-validation strategy, generation checkpoints and the evaluation metrics.

Chapter 4 is the presentation of a system design and experimental work while the result and discussion are presented in chapter 5. Chapter 6 summarizes the key findings, contributions and future research directions. The report is followed by the references and appendices.

CHAPTER 2
LITERATURE REVIEW AND PROBLEM BACKGROUND

2.1 INTRODUCTION

This chapter provides a brief summary of the primary research areas, from which NeuroTrace has been developed. The first one is on hallucinating and lying in Large Language Models (LLMs); e.g., studies of unreliable model behaviour are introduced in benchmarks such as TruthfulQA and HaluEval. The second group is about methodological techniques to find undesirable behaviour like output-based detection, hidden-state and internal representation analysis, safety monitoring and early/pre-generation detection. The studies such as this offer motivation to use more than just the final product; that is, they motivate the use of internal model signals. The review then focuses on the most important focus of NeuroTrace, temporal hidden-state analysis: differences in hidden states can then be investigated across successive tokens generated, to see if useful detection signals emerge in advance of the end of generation. The research gap tackled by the project is stated in Section 2.8.

2.2 HALLUCINATION AND TRUTHFULNESS IN LARGE LANGUAGE MODELS

Hallucination and truthfulness are important aspects of evaluating the reliability of Large Language Models (LLMs). LLMs can generate coherent and fluent answers, but they can also fabricate information, or provide untrue and unsupported information, even information that contradicts known information. Researchers have developed benchmarks specifically focused on the factual reliability of responses generated, to understand this behaviour. Of these, TruthfulQA is a special case that quantifies the accuracy of answers produced by models to questions that are crafted to reveal some of the common misconceptions, and HaluEval provides examples to work through to try to analyze the hallucination-related behaviour of language model answers. The unreliable generation benchmarks provide background information about unreliable generation and motivation for studying other signals, the focal point of NeuroTrace.

2.2.1 TruthfulQA

The goal of the TruthfulQA benchmark is to assess the ability of language models to respond truthfully to questions instead of parroting common misconceptions or inaccurate information. It has questions from different categories to challenge models to avoid providing answers that may appear correct but actually are wrong. The benchmark can be used for analysing the limitations of language models in extracting trustworthy information and can be used as a reference to assess the model behaviour. To rephrase the NeuroTrace context, the quantitative experiments are run in a controlled context based on the HaluEval context, while

the context for NeuroTrace is that of the TruthfulQA is considered as part of the general context of the truthfulness evaluation.

2.2.2 HaluEval

HaluEval is a benchmark to evaluate hallucination in Large Language Models (LLMs). It provides examples that can be used to explore whether the answers generated contain "hallucinated" or unreliable information and it also provides a helpful "ground" to compare the behaviour of the models under different types of answers. HaluEval is used to produce an experimental dataset of pairs of target conditions and correct answer / hallucinated answer questions from NeuroTrace, a balanced set of target conditions and questions with a correct answer and a hallucinated answer. The project is to see if differences can be observed in the model's hidden-state trajectory under controlled conditions, and explore the possibility of observing the difference during generation.

2.3 OUTPUT-BASED HALLUCINATION DETECTION

In the case of output-based hallucination detection, we look to identify unreliable information from the generated text by a language model. Most of these methods work on the final or intermediate text output of the model without having to access the hidden representations within the model. Common approaches to verifying possible hallucinated information involve verifying factual consistency, semantic similarity, contradiction or other characteristics of the generated response.

A direction of research is consistency based detection, multiple responses for the same input, and then comparing the responses. Results should be repeatable over generations (if it is significantly different then this may be a potential indicator of hallucination). These methods can be useful since they can operate with little access to the underlying model and they can include information about the reliability of the generated response.

However, in output-based methods, they are likely to be more concerned with the output of the text than the process of calculating the text. In this way, they can only signal for detection when part of or the whole response is generated, but may not detect changes to the model in the token generation process. This constraint is a motivation to learn methods that involve internal representations and hidden states. In contrast, NeuroTrace considers the temporal evolution of hidden states across the tokens that it generates as well as only the final text.

2.4 INTERNAL REPRESENTATION-BASED DETECTION

Internal representation based detection methods use the internal representation of the information contained in the hidden states of a language model, instead of the information stored in the output. Hidden states are a collection of intermediate numerical representations that are generated along the way by multiple layers in the model during the processing of

tokens. The question has been asked if these representations have informative cues to determine if a model is reliable or unreliable. Features from hidden states may be the activation statistics, distance between activations or the variations in the representation over successive tokens, which are then fed into a separate detection model.

2.4.1 INSIDE

The approach named as INSIDE is based on an internal representation of the model and explores if information from a language model can be used to detect hallucinated responses. The method is based on the internal representations of the model, that it can emit signals beyond the generated text that can be correlated with the reliability of the generated answer. The results of this work show that the potentiality of adopting the internal model state as an extra information source for hallucination detection.

2.5 INTERNAL STATE MONITORING FOR SAFETY

To model representations beyond the detection of hallucinations, to the identification of potentially unsafe or undesirable model behaviour, internal state monitoring is a phenomenon that can be applied. These approaches also tend to assess the final output and signals produced in the model that can contain information about undesirable or dangerous behavior, such as internal states and representations. This direction can be helpful as the internal signals can give information about the behaviour of the model, while the response is still being generated.

There have been recent studies that have looked at hidden state monitoring of behaviours including jail breaks, harmful content and even misbehaviour on other models. They demonstrate that using internal representations is not a requirement of changing the language model itself, but can be used to help monitor model behaviour. NeuroTrace does not only focus on this general monitoring strategy but narrows it down to the temporal evolution of hidden state changes during generation, which could reveal a change in the hidden state, as an early detection signal, in addition to the generated output.

2.5.1 HiddenDetect

HiddenDetect is a study that explores the use of Hidden State Representations for detecting jailbreak-related behavior within LLMs. The method does not only consider the output, but also the internal representation of the model and checks if the model contains unsafe or undesirable input signals. This work illustrates the use of hidden states to provide useful information for tracking model behaviour and as an additional source of evidence for identifying potentially harmful behaviour.

2.5.2 SIREN

In Large Language Models, SIREN will examine the ability of internal representations to monitor potentially harmful/undesirable behaviors. The approach is on the information contained within the model and not the output, such as the model's internal state. The overall idea is that hidden representations could be helpful for conveying information about the behaviour of the model and could be employed to monitor the model in addition.

2.6 EARLY AND PRE-GENERATION DETECTION

Methods used earlier or pre-generation detection can be employed in order to determine if a potentially problematic model behavior is taking place prior to the full response. This direction is significant because the ability to intervene, stop generation or take corrective action in a system might be limited after generation. Recent studies have thus examined if pre - or in-generation information can also be useful information in predicting the behaviour of a model.

2.7 TEMPORAL HIDDEN-STATE ANALYSIS

The changes of the model representations, as the model processes and generates tokens, is the focus of temporal hidden-state analysis. This view considers the hidden state of a response as a chain of hidden states that is created at each step of generation, instead of considering the hidden state of a single response. Various measurements can be used to represent changes between these states, such as vector distance, cosine similarity, and relative statistic changes in activation. This will provide a chance to explore the internal route through the model used during generation.

2.8 RESEARCH GAP AND PROBLEM BACKGROUND

Although it is known that the outputs and internal representations of generated performances can be useful in recognizing undesirable or unreliable LLM behaviour, it is unclear what happens to these internal signals during the course of the generated performance. Most current approaches study the output (target) or analyze internal representations, but do not explicitly consider the temporal dynamics of the hidden states between consecutive tokens. This is because it is difficult to determine what the detection signal will be when it is useful, and how much of the response has yet to be generated after the detection signal.

However, NeuroTrace is able to overcome this limitation by analyzing the temporal trajectories of hidden states at the level of tokens from a frozen GPT-2-medium model. The project doesn't make any changes to the language model or fine-tune the language model, it just uses the intermediate layers to extract representations and use those representations to get the trajectory and uncertainty features. The performance of detection is evaluated at a number of checkpoints corresponding to 25%, 50%, 75% and 100% of the entire wave period, and in addition to the detection performance, the time of the detection signal is also studied. The

project explores therefore if temporal changes in internal representations can give an early indication of potentially anomalous generation and if this information can be leveraged using a light-weight detection model.

2.9 CHAPTER SUMMARY

The chapter examined the current important research directions of detecting unreliable and undesirable behavior in LLM. It contained a set of benchmarks for hallucination and truthfulness as well as output-based detection methods and the internal model representation-based methods. It also involved an internal-state monitoring for safety, early and pre-generation detection and temporal analysis of hidden states. The reviewed literature revealed that there was a gap in the literature regarding the evolution of a hidden-state representation over time during the generation of tokens, particularly when the detection signal is useful. The lack of this has motivated NeuroTrace, which proposes to investigate temporal hidden-state trajectories of a frozen GPT-2-medium model with various generation checkpoints and a lightweight detection mechanism.

CHAPTER 3
PROBLEM STATEMENT AND METHODOLOGY

3.1 PROBLEM STATEMENT

Large Language Models (LLMs) can generate coherent responses that contain errors, misinformation or even that are completely missing. Most detection methods are based on the final product, and cannot detect problematic behavior early. To tackle this challenge, NeuroTrace examines the potential of internal hidden-states representation of a frozen GPT-2-medium model to serve as a helpful signal for detecting possible unusual generation during the token-wise generation process. It conducts a study to uncover hidden states from intermediate layers, creates trajectory and uncertainty features and utilizes a light-weight Logistic Regression detector for classification of controlled generation conditions.

The problem is formulated as an early-warning detection problem. The sequence generated is not evaluated for the entire response but evaluated at 25%, 50%, 75% and 100% of the response. The aim is to identify if there are any possible internal hidden-state trajectories which can distinguish the controlled target conditions and, most importantly, if a detection signal can occur before the generation. The study also provides quantitative measures of detectability of the tokens, and how many tokens remain after the first warning to indicate potential lead time that is offered by the proposed approach.

3.2 OBJECTIVES

Every attempt of NeuroTrace is to investigate whether such useful signals can be acquired from the internal hidden-state representations of a frozen model of GPT-2-medium that may aid the early detection of potentially anomalous generation. To observe hidden states throughout token-by-token token generation, to extract temporal trajectory and uncertainty based features and to feed these features to a lightweight Logistic Regression detector. It also aims to evaluate detection performance at 25%, 50%, 75%, and 100% token generation points, determine the delay to the first detection signal, estimate the number of remaining tokens after the first detection signal, compare the performance of various sets of features (trajectory-based vs. uncertainty-based), and implement group-aware cross validation to reduce data leakage. In general, the project aims to explore the possibility of temporal hidden state trajectories as an early warning signal without any changes and fine-tuning of the underlying language model.

3.3 DATASET

HaluEval is the main dataset employed in NeuroTrace, tailored to facilitate the investigation of hallucination related behaviour in LLMs. The official HaluEval qa_data.json file contains 10,000 records, each having a knowledge, question, right_answer, and hallucinated_answer. For the controlled experiment, 100 questions were selected and a set of

100 conditions of the questions were set up with the correct answer and a set of 100 conditions with the hallucinated answer. This resulted in an equal number of 200 controlled samples, and 100 samples for each target condition. The project also tested sets of related interests like ‘TruthfulQA’ and ‘XSTest’, but did not use these for the final quantitative evaluation of the detector.

3.4 DATA PREPARATION AND CONTROLLED CONDITIONS

The HaluEval data was reformatted to a controlled experiment for inspecting the internal working of the frozen GPT-2-medium model. We randomly selected 100 questions, each of which makes 2 conditions: the right-answer (target label 0) and the hallucinated-answer (target label 1) condition, for a total of 200 balanced samples (100 per target class). The samples are then related in the sense that they have a common sample identifier, and this will be the grouping variable for the cross validation.

3.5 DATASET CONSTRUCTION

The final experimental data was derived from scaling data for the selected HaluEval questions to two controlled conditions, per question. The first condition is a pair of a question and the right answer to it; the second condition is a pair of a question and the hallucinated answer to it. Two conditions are evaluated as a pair, with each pair given a common sample ID to ensure the correlation between the two conditions in the evaluation. This process yields 200 samples from 100 different questions, such that 100 samples are from the first target class and 100 from the second. The samples that are generated by the construction are passed to the frozen GPT-2-medium for the generation of the tokens, and then passed to the hidden-state and uncertainty information to build the features. The paired conditions also enable group-aware evaluation – the groups of questions are kept together in the same fold to minimize data leakage.

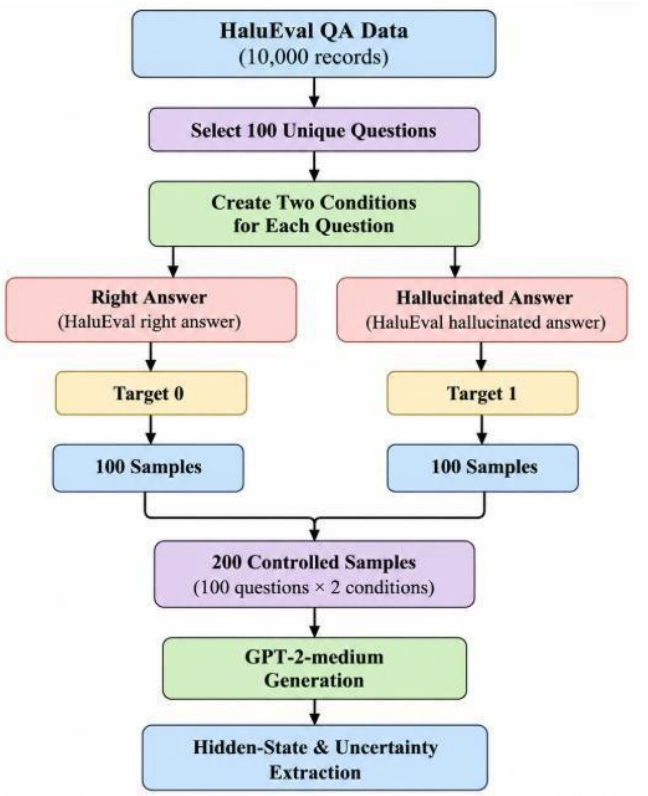


Figure 3.1: HaluEval Dataset Construction and Controlled Sample Generation

3.6 HIDDEN-STATE FEATURE TYPES

NeuroTrace leverages both features from the hidden representations and output distributions of the frozen GPT-2-medium model. The hidden states in layers 3, 4, 5 and 6 are gathered and studied at the following generation steps for each generated token. The extracted features are the change in strength and direction of the internal representation of the model and the uncertainty at the token level. These features are then integrated together to form response level statistics and used as input to the detection model. The key feature types used in the project are hidden-state norm, hidden-state transition, cosine similarity, relative state change, token likelihood and entropy.

3.6.1 Hidden-State Norm

Hidden-state norm is the size of the hidden-state vector of a specific layer on a generated token. The L2 norm of a hidden-state vector h_t^((L)) at token t and layer L is:

$$\left\| h_t^{(L)} \right\|_2 = \sqrt{\sum_{i=1}^{d} \left(h_{t,i}^{(L)}\right)^2}$$

where d is the dimension of the hidden state. NeuroTrace is based on the 1024 hidden-state size of the GPT-2-medium model. One of the features used for analysing the changes in the internal state of the model is the norm which is a measure of the overall size of the internal representation at each generation step.

3.6.2 Hidden-State Transition

The state transition of the hidden state is the difference between the state of the model at two consecutive steps of the sequence of generated tokens. Current hidden-state vector h_t^((L)) and previous hidden-state vector h_t-1^((L)) are compared and transition (between hidden-state vectors at t and at t-1 at layer L) is calculated as the Euclidean distance between h_t^((L)) and h_t-1^((L)) .

$$\Delta_t^{(L)} = \|h_t^{(L)} - h_{t-1}^{(L)}\|_2$$

The larger the transition value the larger the difference in the underlying representation at each generation step; the smaller the transition value, the closer the representations are. In NeuroTrace, they are calculated across 4 layers (3, 4, 5 and 6) and used as features for temporal trajectories. The sequence of the various transitions between tokens provides an indication of the state of the model's internal state as the model evolves through its state transitions.

3.6.3 Cosine Similarity

Cosine similarity is used to quantify the similarity of the direction of the two hidden-state representations of consecutive generated tokens. The ratio of the current and previous hidden-state vectors normalized by the magnitudes of the current and previous hidden-state vectors:

$$\text{CosSim}_t^{(L)} = \frac{h_t^{(L)} \cdot h_{t-1}^{(L)}}{\|h_t^{(L)}\|_2 \|h_{t-1}^{(L)}\|_2}$$

The value is usually between −1 and 1: positive values indicate that the two representations go in the same direction, and negative values indicate in the opposite direction. In NeuroTrace , cosine similarity is calculated between the next token generated in the 3rd, 4th, 5th and 6th layers. It is used in conjunction with the transition feature of the internal representations to record the direction of motion between the two internal representations that will follow one another, but not just the distance.

3.6.4 Relative State Change

Relative state change: state change from the previous hidden-state change, according to the size of the previous hidden-state representation. It can be calculated as the Euclidean distance from the previous hidden state to the next hidden state divided by the norm of the previous hidden state:

$$\text{RelativeDelta}_t^{(L)} = \frac{\|h_t^{(L)} - h_{t-1}^{(L)}\|_2}{\|h_{t-1}^{(L)}\|_2 + \epsilon}$$

where ϵ is a small number to avoid a divide by zero. This feature returns a normalized value which is proportional to the size of the current state transition, relative to the size of the previous

state. Relative state change is calculated for layers 3, 4, 5 and 6 for the NeuroTrace model and represents the strength of the temporal change in the internal representations of the model.The probability and entropy of the tokens are computed.

3.6.5 Token Probability and Entropy

The probability of the token is that the token has when the agent decides to produce it, as computed by GPT-2-medium. It provides the trust/confidence in the model's prediction at the token level. The entropy of the model's probability distribution over the possible next tokens is the uncertainty. The entropy of a probability distribution p_i is given by:

$$H = -\sum_{i=1}^{V} p_i \log p_i$$

The higher the entropy, the more spread out the probability distribution and the more uncertainty there is, the lower the entropy value the more probability mass on fewer tokens. In generating the tokens, the probability of each token and entropy are extracted to be uncertainty features added to the hidden-state trajectory features in NeuroTrace . These features allow comparison of the information provided by uncertainty information and the information provided by the temporal hidden-state representations in the experiments.

3.6 FROZEN GPT-2-MEDIUM MODEL

NeuroTrace uses GPT-2-medium as the language model to generate text and to extract hidden-states. It contains approximately 354.8M parameters, and 1024 hidden states. For this project, the model is frozen (all parameters are fixed in the process of feature extraction and training the detector). All hidden representations at layers 3, 4, 5 and 6 are collected for each token created. This setup allows the project to obtain knowledge on the inner working of a pre-trained deep learning model without modifying the model's parameters or fine-tuning. The extracted representations are then converted to temporal trajectory features, which are then fed into the lightweight detection model.

3.7 TOKEN-LEVEL TRAJECTORY EXTRACTION

NeuroTrace keeps the hidden-state representations produced by GPT-2-medium for each token that it has generated through the course of its generation. For layers 3, 4, 5 and 6 the hidden state of each layer, together with the token probability and entropy are saved for each token. These hidden states are then compared to the next generation steps to measure the distance of the transition, the cosine similarity and the relative change of states. This gives a series of measurements of the internal representation of the model over time in the course of the generation process. Token level data retrieved and stored for other feature aggregation / early-warning analysis through checkpoints.

3.8 FEATURE AGGREGATION

The features at the response level are created by aggregating token-level features in the generation process; the entire generation can be represented by a fixed-length feature vector. A set of statistical quantities like mean, std deviation, max, min are calculated for each trajectory and uncertainty feature, based on the tokens produced. As an additional option, it is also mentioned how many tokens were generated. This process combines the sequence of variable length tokens into a fixed length representation that will be fed into the detection model for processing.

This yields a 53 response level feature column, including the token count feature. These aggregated features are a mixture of the information regarding the magnitude and temporal changes of the hidden states as well as the model's token-level uncertainty. This conglomerated representation is then passed on to the Logistic Regression detector as well as to the next ablations and early-warning experiments.

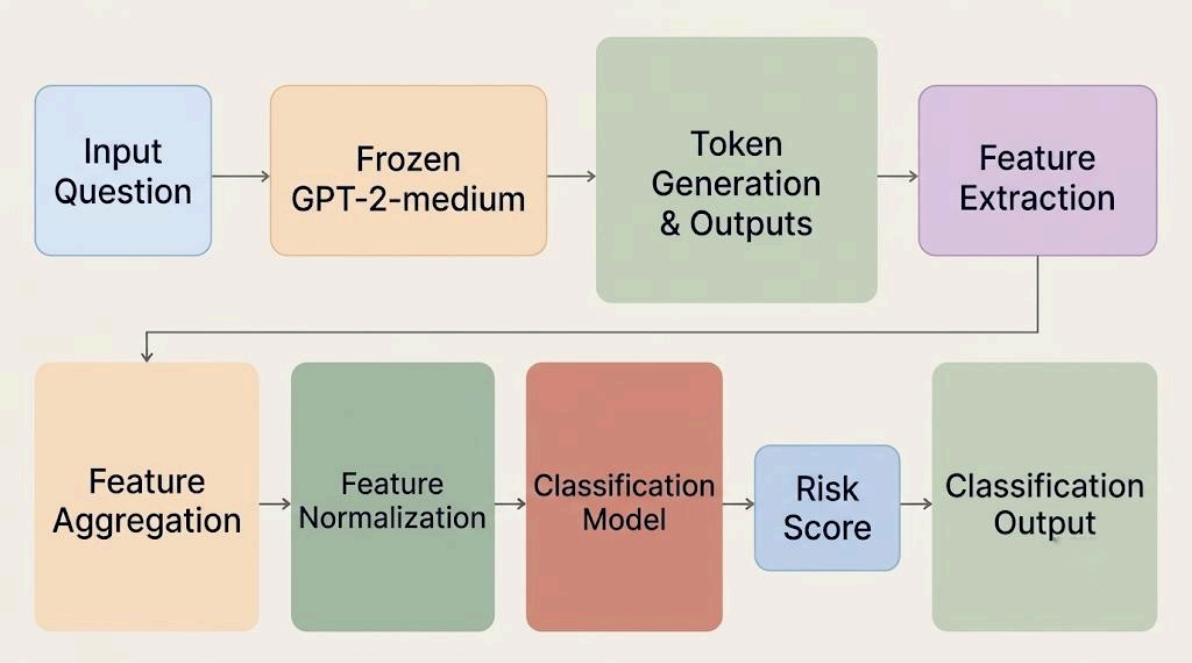


Figure 3.2: NeuroTrace Feature Extraction and Methodology Flowchart

3.9 LOGISTIC REGRESSION DETECTOR

The classification model used for the detection of the controlled generation conditions is a lightweight model, namely Logistic Regression named neuroTrace. StandardScaler is applied to the feature vectors to standardize the features so that they are scaled in a similar manner, in case they have different numerical ranges. The outcome of these standardised features is then sent to the Logistic Regression Classifier, that gives a score that represents the likelihood of the sample being within the target condition. The experiments done for detection purposes have

been using a fixed decision threshold value of 0.5 for making the final classification decision between model outputs of final NORMAL or SUSPICIOUS.

Logistic Regression is used for the detection component for simplicity and cheap computational cost. A complex classifier is not required since the aim of NeuroTrace is to determine if any of the information in the hidden-state of the temporal features is useful information for detecting. The detector is thus a downstream analysis module, with the main information being provided by the hidden state trajectories which are extracted from the frozen GPT-2-medium model.

Parameter	Configuration
Classifier	Logistic Regression
Feature Scaling	StandardScaler
Input Features	Aggregated hidden-state and uncertainty features
Trajectory-only Features	40
Total Aggregated Features	53
Classes	Target 0 and Target 1
Decision Threshold	0.5
Language Model	Frozen GPT-2-medium

Table 3.1: Logistic Regression Detector Configuration

3.10 GROUP-AWARE CROSS-VALIDATION

To reduce the possibility of data leakage during model evaluation, NeuroTrace uses StratifiedGroupKFold cross-validation. In the controlled data, there are two conditions that are derived from the same question, and the correct answer and the hallucinated answer. These two samples are alike and the training and testing sets might induce the detector to learn indirectly the information specific to the question. The configuration is StratifiedGroupKFold with 5 groups, the ID of the "sample" is the grouping variable, and the "target" class is used for stratification, with 200 samples and 100 unique groups of questions.

To prevent this, two conditions of the same question are assigned the same sample ID and this is the variable cross-validated between the two conditions. StratifiedGroupKFold maintains class balance, and also maintains the samples from the same question in the same fold. This kind of assessment approach provides a closer estimate of the discriminating power of the features extracted to differentiate the two controlled conditions on the test sets of question group types not seen before.

3.11 EARLY-WARNING CHECKPOINTS

To NeuroTrace analyzes the generated sequence at 4 different points: 25%, 50%, 75%, and 100% of generated tokens. The trajectory features are extracted and summed up as they are seen up to each of the trajectory checkpoints rather than until all of the reply has been received. This will allow the detector to create an intermediate risk score and classification of the risk while generating.

It is important to measure the timing of the detection as well as classification performance for the checkpoint-based approach. The performance of the detector is assessed for each of the checkpoints using ROC-AUC and other classification metrics. The number of tokens that are still not detected after the first warning is also noted to determine the potential detection lead time. The four checkpoints are the first quarter, half, three-quarters and half and full (final detection).

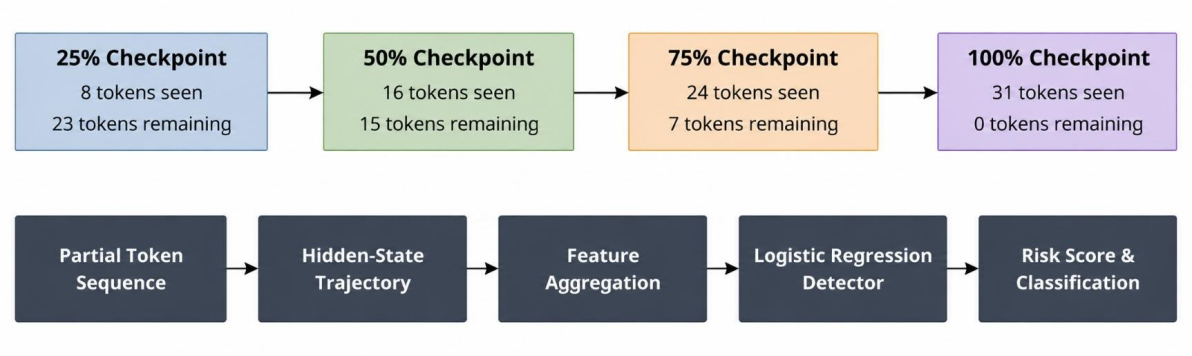


Figure 3.3: NeuroTrace Early-Warning Checkpoint Workflow

3.12 RISK SCORE AND CLASSIFICATION

The trained Logistic Regression detector returns a probability score for every input sample that is used as a risk score in NeuroTrace. The score represents an estimate of the likelihood that the features came from the target condition, based on the extracted features. Please note, this score is generated from the model for controlled experimental conditions, and does not provide a probability of fact hallucination.

A risk score of 0.5 is used to classify. Samples with a score < 0.5 are considered NORMAL and those >= 0.5 are considered SUSPICIOUS. The threshold is used throughout the evaluation of the performance of the detection, to ensure uniformity in comparison of the different stages of generation.

3.13 DETECTION LEAD TIME

The time between the NeuroTrace detection of a target-1 controlled condition and the detection of that condition by NeuroTrace, before the generation process is completed is the detection lead time. For each sample the detector is tested at the quarter, half, 3/4 and full

points. If the risk score hits the classification level, the first point of occurrence is noted as the first warning point.

The amount of tokens left after this first warning is then used to determine the amount of available lead time. The more tokens there are the more advance warning the detector provided during generation. Samples that do not end up as suspicious are kept distinct. With this measure, NeuroTrace can not only determine if the controlled condition can be detected, it can also determine the time of the detection signal.

Measure	Description
First warning point	Earliest checkpoint classified as SUSPICIOUS
Tokens remaining	Tokens left after the first warning
Mean lead time	Average tokens remaining across detected samples
Median lead time	Median tokens remaining
Detection rate	Percentage of target-1 samples detected by generation end

Table 3.2: Detection Lead-Time Measures

3.14 ABLATION STUDY

Ablation studies are used to determine which types of features are useful to the task at hand. NeuroTrace uses three sets of features, namely trajectory only features, uncertainty only features, and all of the extracted features. The features used in the trajectory-only configuration contain hidden-state norms, transitions, cosine similarities and relative state changes; and in the uncertainty-only configuration, token probability and entropy features are employed. All-feature is a combination of both groups.

The same group aware cross validation procedure is used for evaluation of each configuration, ensuring the evaluation is done under a consistent set of conditions. Temporal hidden-state trajectory information is then compared to the ROC-AUCs of the various configurations of the feature set to determine if there is additional value in detecting the presence of temporal hidden-state trajectory information beyond the token-level uncertainty information.

The feature configurations considered in the ablation study are summarized in Table 3.3.

Feature Configuration	Features Included
Trajectory-only	Hidden-state norm, transition, cosine similarity, relative state change
Uncertainty-only	Token probability, entropy
All features	Trajectory + uncertainty features

Table 3.3: Feature Configurations Used in Ablation Study

CHAPTER 4
SYSTEM DESIGN AND EXPERIMENTAL WORK

4.1 OVERALL SYSTEM ARCHITECTURE

The NeuroTrace system aims to be a sequential pipeline, which follows the inner workings of a frozen GPT-2-medium model for text generation. A control input vector sample is given to the system for starting, and the output is produced token-by-token while the hidden-state representations are collected from layers 3, 4, 5 and 6. In the meantime, probability and entropy values of the tokens are retrieved to show uncertainty in the generation. These signals are then translated into trajectory and uncertainty features, for further analysis.

The features are extracted, aggregated to a fixed length representation, standardized and given as input to a Logistic Regression detector, which has been trained on the extracted features. The detector has a risk value assigned and is classified as NORMAL or SUSPICIOUS by using a threshold value of 0.5. The same pipeline is used at the 25%, 50%, 75% and 100% generation checkpoints for early-warning analysis. Overall architecture of NeuroTrace system is shown in Figure 4.1.

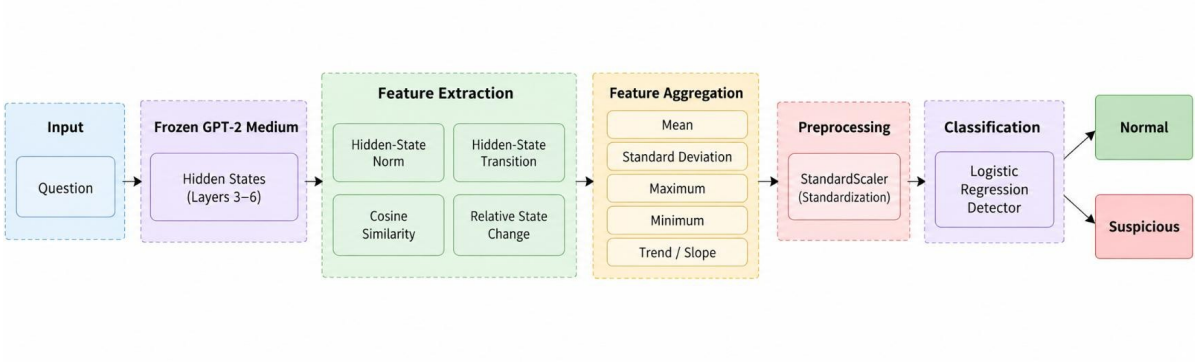


Figure 4.1: Overall Architecture of the NeuroTrace Detection System

4.2 DATA FLOW

The NeuroTrace data flow refers to the process of a controlled input being processed from the initial stimulus to when the final detection result is produced. The input is passed into the frozen GPT-2-medium model and then the GPT-2-medium's output is generated token-wise. In this procedure, the hidden-state representational information of layers 3, 4, 5 and 6 are gathered along with tokens and their probabilities and entropy values. The temporal trajectory and uncertainty features are constructed using these token-level features.

The extracted features are then passed to a fixed-length representation (feature vector), and scaled using StandardScaler. The standardized representation is passed to the trained Logistic Regression detector who gives a risk score. The sample is declared NORMAL or

SUSPICIOUS on the basis of the fixed value 0.5. Early-warning analysis is performed for the same data flow at the 25%, 50%, 75% and 100% generation levels.

4.3 MODULE DESIGN

The NeuroTrace system can be split up into a series of functional modules, all of which execute a specific part of detection. Modular design Data Preparation, Model Generation, Hidden State Extraction, Feature Processing, Detection and Early Warning Analysis. This allows for an easier system understanding, evaluation and extension without altering the frozen language model.

The major modules of NeuroTrace are: Dataset Preparation Module, Frozen LLM Generation Module, Hidden-State Extraction Module, Temporal Trajectory Module, Uncertainty Feature Module, Feature Aggregation Module, NeuroTrace Detection Module, Early-Warning Module and Risk Score Generation Module. The overall architecture is shown in Figure 4.1 and these modules are connected as such. Requires no special explanation because an overall system architecture is provided which demonstrates the relationships between modules.

4.4 DATASET PREPARATION MODULE

The Dataset Preparation Module is used to help transform the original data from HaluEval to the controlled experimental format needed by NeuroTrace. The module begins with the official question-answer records of the HaluEval; 100 different questions are selected for the main experiment. For the questions chosen, two conditions are established; one condition contains the correct answer to the question, and another condition contains the hallucinated answer to the question. This is a balanced data set with 200 samples evenly divided into 2 target conditions—100 for each.

If the two conditions of the same question are different, the same sample ID is used for evaluation for each pair of samples. The prepared samples are then fed into GPT-2-medium generation module, and the uncertainty information of each token and hidden state are extracted from the module. It thus provides a consistent structure of the input for feature extraction, group-aware cross-validation, ablation analysis and early-warning evaluation.

Construction and production of controlled samples is detailed in Section 3.5 and shown in Figure 3.1. In this case, the focus is on the significance of the preprepared data for the bigger NeuroTrace pipeline.

4.5 FROZEN LLM GENERATION MODULE

The Frozen LLM Generation Module is in charge of generating responses from the prepared input samples with the help of GPT-2-medium. No fine-tuning or parameter changes

were made to the model to ensure that the experiments can be conducted under well-controlled conditions of generation. During the generation phase, the model outputs the response token one token at a time and the information for the input and output level is written down at each generation step.

NeuroTrace keeps hidden-state representations on layers 3, 4, 5, and 6 of the model, along with token probabilities and entropies per token generated. A maximum of 32 new tokens are used for the generation process, the temperature is 0.7 and the top-p value is 0.9. The information collected is sent to HSE and feature-processing modules, which produce the temporal trajectory feature and uncertainty feature.

4.6 HIDDEN-STATE EXTRACTION MODULE

The Hidden-State Extraction Module is used to extract internal representations produced by GPT-2-medium for each token produced. The hidden-state outputs from layers 3, 4, 5 and 6 are dropped and stored for each token produced for use later. In GPT-2-medium, there are 1024 hidden states, so each hidden state extracted will be the internal information that the model has processed at each generation step.

The hidden states are extracted and analyzed throughout the generation to gain insights into the model's internal representation. The hidden-state transitions are calculated for each pair of consecutive hidden states, cosine similarities, relative state changes are calculated, and the hidden-state norms contain size information about the hidden-state representations. These measurements are used to determine the temporal trajectory that is used by the NeuroTrace detector.

The extraction process is done without changing the parameters of GPT-2-medium. This means that the module is just an observer of the language model, and the internal representation of the frozen language model does not affect the behaviour of the original language model.

4.7 TEMPORAL TRAJECTORY MODULE

The Temporal Trajectory Module is a module which quantifies the inner dynamics of the hidden representations of GPT-2-medium over successive generated tokens. NeuroTrace considers each hidden state as one occurrence of the state, but sees the sequence of hidden states as a model's internal temporal dynamics that will follow the way that the model's internal representation evolves during generation.

For each layer being monitored, the Euclidean transition distance and cosine similarity and relative state change are calculated between successive hidden states. These measurements include the speed of change as well as the direction of change between generations. The magnitude of the internal representations are also remembered in the hidden state. The resulting measurements are then used to construct the trajectory feature set of the downstream detector.

The trajectory information will be extracted during the generation process, and can also be cut at the points of 25%, 50%, 75% and 100%. This enables the same trajectory features to be assessed at various points in generation and allows for an early-warning analysis done by NeuroTrace.

4.8 UNCERTAINTY FEATURE MODULE

The Uncertainty Feature Module is a module that records the uncertainty of tokens throughout the generation process of GPT-2-medium. It stores the probability of the generated token, the entropy of the output probability distribution, and the normalized entropy which together represent the level of confidence of the model in generating the token. The values are complementary to the hidden-state trajectory features in the sense that they inform about the behaviour of the output level of the model instead of its internal representations. They are summarized over the tokens available up to the selected point in the trajectory and used in the uncertainty-only and all-feature versions of the ablation study.

4.9 EXPERIMENTAL SETUP

The experimental setup describes the setup configuration of the NeuroTrace framework implementation and testing. The frozen language model used for the experiments was GPT-2-medium and the hidden state representations were extracted from the third, fourth, fifth and sixth layers of the model. It was used to generate new tokens, one by one, with a maximum of 32 new tokens, a temperature of 0.7 and a top_p value of 0.9.

The collected data set consisted of 200 controlled samples from 100 different questions in HaluEval. We extracted features, scaled them using StandardScaler and tested them with a Logistic Regression detector. Group-aware StratifiedGroupKFold was used for the evaluation, using question/sample ID as the grouping variable. Experiments with the early-warning were run at 100%, 75%, 50%, and 25% of the sequence generated.

The experimental setup for implementation of the NeuroTrace is summarized below.

Parameter	Configuration
Language Model	GPT-2-medium
Model Parameters	354,823,168
Model Status	Frozen
Hidden Dimension	1024
Monitored Layers	3, 4, 5, 6
Dataset	HaluEval
Controlled Samples	200
Unique Questions	100
Maximum New Tokens	32
Temperature	0.7
Top-p	0.9
Detector	Logistic Regression
Feature Scaling	StandardScaler
Cross-Validation	StratifiedGroupKFold
Number of Folds	5
Early-Warning Checkpoints	25%, 50%, 75%, 100%
Classification Threshold	0.5

Table 4.1: Experimental Configuration

4.10 HARDWARE AND SOFTWARE REQUIREMENTS

The experiments with NeuroTrace were done with a GPU computational environment to generate and extract hidden states from GPT-2-medium. Experiments were run with a Tesla T4 GPU and CUDA acceleration. This was especially helpful in the case of GPU execution, as the hidden-state representations needed to be gathered for several layers over successive generated tokens.

It is applied to the software world (Python and PyTorch) , and is loaded and executed with GPT-2-medium, provided by Hugging Face Transformers library. The techniques that were used for feature scaling, logistic regression, cross validation and evaluation were scikit-learn. The primary version of software used for the experiments is shown in Table 4.2.

Component	Configuration
GPU	NVIDIA Tesla T4
GPU Acceleration	CUDA
Deep Learning Framework	PyTorch 2.11.0+cu128
Transformers	5.16.1
Scikit-learn	1.6.1
NumPy	2.1.3
Pandas	2.2.3
Programming Language	Python
Language Model	GPT-2-medium

Table 4.2: Hardware and Software Configuration

CHAPTER 5
RESULTS AND DISCUSSIONS

5.1 EXPERIMENTAL RESULTS

The NeuroTrace experiments were run to see if can useful signals be learned from temporal representations of hidden states of a frozen GPT-2-medium model that could help differentiate between the two controlled generation conditions. The evaluation of three main aspects is the contribution of different feature types, the performance of the detector at different steps in its generation and the timing of the first warning detection.

Group-aware cross-validation is used to compare trajectory-only, uncertainty-only and all-feature configurations. In addition, the early-warning experiment also tests the trajectory-based detector at 25%, 50%, 75% and 100% generation points. Besides ROC-AUC, two other metrics for classification performance and detection lead time are introduced to provide a more complete description of the performance of the detector.

The results show that the temporal hidden-state trajectory features have a stronger signal than uncertainty-only features in the experimental controlled setting. The best performance at an early stage was achieved by the trajectory-only configuration with a mean ROC-AUC of 0.9095 at the 25% checkpoint. The results also demonstrate that the detection performance is different for each generation stage, meaning that the internal signal is not the same at the end of the complete response. The dataset and experimental setup for these exp eriments are summarized in Table 4.1; 31 usable generated tokens (out of a possible 32) were available for each sample in the controlled setup.

5.2 ABLATION STUDY RESULTS

The different types of features of NeuroTrace were analysed using the ablation method. Three different setups were attempted using the same 5-fold group-aware cross-validation: only trajectory features, only uncertainty features, and features from both trajectory and uncertainty. This comparison helps to show if there is temporal level uncertainty information that complements the token level uncertainty information.

The trajectory-only configuration was the best with a mean ROC-AUC of 0.7700, a SD of ±0.0417, and a 95% CI of [0.7335, 0.8065]. The results for the uncertainty-only and all-feature types of the ROC-AUCs were 0.6030 and 0.7500 respectively. The discriminative power of these results indicates that the temporal hidden-state trajectory features were more discriminative than the uncertainty features, and alone in the controlled experiment.

The performance of trajectory-only configuration was slightly better than the uncertainty-only configuration (ROC-AUC = 0.167). It was also marginally better than the all features configuration (0.020 ROC-AUC). In this experiment, this suggests that the uncertainty

features did not have a significant positive impact on the overall group-aware ROC-AUC, but the uncertainty information can serve as additional information at certain checkpoints.

The results of the ablation are summarized in Table 5.1.

Feature Set	Mean ROC-AUC	Standard Deviation	95% Confidence Interval
Trajectory-only	0.7700	±0.0417	[0.7335, 0.8065]
Uncertainty-only	0.6030	±0.0402	[0.5678, 0.6382]
All features	0.7500	±0.0499	[0.7063, 0.7937]

Table 5.1: Group-Aware Ablation Study Results

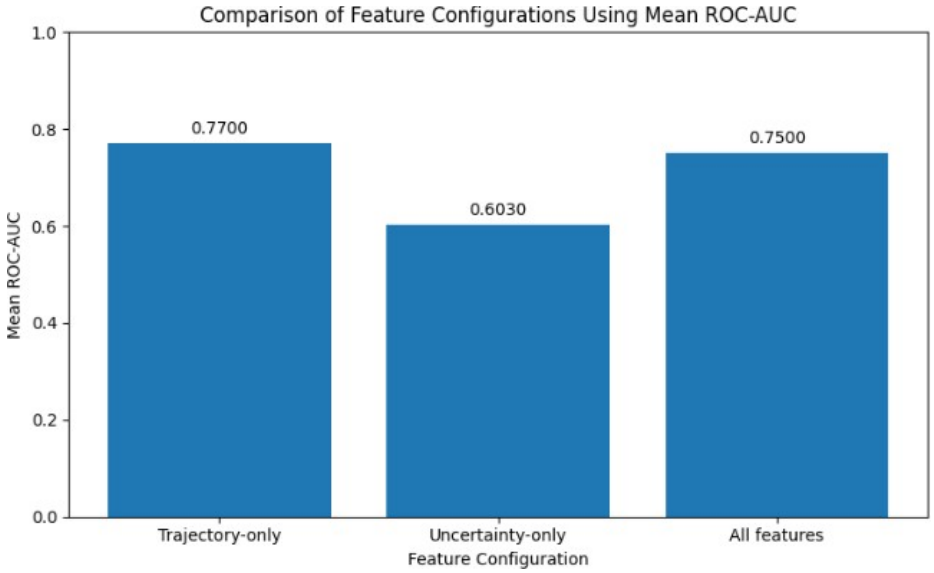


Figure 5.1: Comparison of Feature Configurations Using Mean ROC-AUC

5.3 TRAJECTORY-ONLY RESULTS

In the trajectory-only setting, all the features are extracted from the temporal hidden-states of GPT-2-medium. The following features are hidden-state norms, hidden-state transitions, cosine similarities and relative state changes in layers 3, 4, 5 and 6. This experiment aims to test the effectiveness of the model's internal representations in differentiating between the two controlled target conditions in a trial where they are not based on token probability or entropy features.

The group-aware 5-fold cross-validation setting resulted in a mean ROC-AUC of 0.7700, a standard deviation of ±0.0417 and a 95% confidence interval of [0.7335, 0.8065] for the trajectory-only configuration. This result shows that the hidden-states of the temporal model give a significant discriminative signal in the controlled experimental setup. The trajectory-only is considered with 40 features and the results obtained are compared with the other configurations in Table 5.1.

In particular, the detector uses only the information obtained from the hidden trajectory of the model. The language model is not modified and the detection part can only be a simple Logistic Regression classifier. Thus, the observed performance indicates that there is information in the temporal changes of intermediate representations which could be used by a detector in the downstream module.

5.4 UNCERTAINTY-ONLY RESULTS

The uncertainty-only configuration is a configuration that does not include the temporal hidden-state trajectory features and relies on token probability and entropy features. This experiment was conducted to see if the uncertainty information at the token level alone from the model is useful for distinguishing between the two controlled target conditions.

The uncertainty-only configuration's performance, using the same 5-fold StratifiedGroupKFold evaluation procedure, was a mean ROC-AUC of 0.6030, with a standard deviation of ±0.0402, and a 95% confidence interval of [0.5678, 0.6382]. This performance is less than the mean ROC-AUC that was achieved using only the trajectory features (0.7700).

The comparison suggests that the uncertainty information at the token level was less discriminative than the features at the hidden states over time in the controlled experiment. This result confirms that we can consider the internal representation dynamics as the most important feature source for NeuroTrace, and that uncertainty features can be seen as additional information. The comparison of all three configurations is given in Table 5.1.

5.5 EARLY-WARNING RESULTS

The early-warning experiment compares the performance of the trajectory-only detector for the sequence generated up to 25%, 50%, 75% and 100% of the generated sequence. Only tokens available up to the current checkpoint are used to calculate the aggregated features, which makes the detector to calculate a risk score before the complete response is generated. The results with the same group-aware 5-fold cross validation are displayed in Table 5.2.

Checkpoint	Mean ROC-AUC	Standard Deviation	95% Confidence Interval
25%	0.9095	±0.0618	[0.8553, 0.9637]
50%	0.7625	±0.1069	[0.6688, 0.8562]
75%	0.7090	±0.0933	[0.6272, 0.7908]
100%	0.7030	±0.0938	[0.6208, 0.7852]

Table 5.2: Trajectory-Only Early-Warning Results

For the controlled generation configuration, about 8 tokens were seen at the 25% mark and about 23 tokens are left. Thus, the high ROC-AUC at this point suggests that useful

information will be found in the trajectory relatively early in the generation process. This is the result of the early discrimination of target conditions (controlled) rather than the result of early discovery of the factual hallucination.

A line graph of mean ROC-AUC by generation can be used to visualize the checkpoint-wise performance.

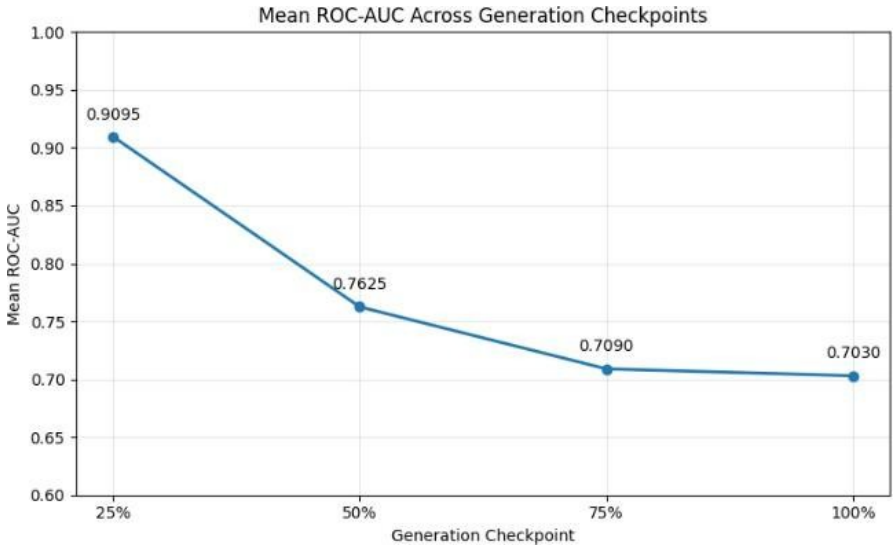


Figure 5.2: Mean ROC-AUC Across Generation Checkpoints

5.6 CHECKPOINT-WISE ROC-AUC ANALYSIS

Considering the detector analysis as a checkpoint-wise analysis, the discriminative performance of the NeuroTrace detector is evaluated with the increase in the number of tokens generated. The evaluation has been performed for a feature configuration for trajectory only, at 25%, 50%, 75% and 100% of the generated sequence.

The best results were found at the 25% checkpoint with a mean ROC-AUC of 0.9095. Around 8 tokens were observed at this stage and around 23 tokens were still in the controlled generation setup. The ROC-AUC decreased to 0.7625 at 50%, 0.7090 at 75%, and 0.7030 at 100%. Thus, there is no monotonic rise in the detection performance with the increasing number of tokens generated.

This behaviour suggests that there could be useful information in the internal trajectory early in the generation process, however, the discriminative signal may change along the generation process. The ROC-AUCs also diminish with time, indicating that there is no systematic increase in separation between the two controlled conditions with additional tokens generated at later times.

The mean ROC-AUC values at the 25%, 50%, 75% and 100% checkpoints are shown in Table 5.2; 23, 15, 7 and 0 tokens were observed at each of these checkpoints, respectively.

The trend in Figure 5.2 gives a visual comparison of the ROC-AUC values for the four checkpoints. This is especially meaningful to the early-warning goal as it is the best discrimination measure, while only a fraction of the response has occurred at 25%.

5.7 DETECTION RATE AND LEAD-TIME ANALYSIS

The detection-rate and lead-time analysis takes into account how often the target-1 controlled condition is detected by the NeuroTrace detector as well as how long it takes between the occurrence of the condition and the first warning triggered. If a fixed threshold of 0.5 is used for the classification of the detector at the points that are evaluated, the classification of the detector is labeled SUSPICIOUS, and the sample is classified as "detected". At the 25% checkpoint, 82 samples were detected out of 100 samples that were detected of the target-1. By 50% to 86 samples, 75% to 88 samples, and by end of generation to 91 samples. All the target-1 controlled-condition samples were measured at the end of generation and the percentage of samples found was 91% below the fixed threshold.

82 samples were issued their first warning at 25%, 4 at 50%, 2 at 75% and 3 at 100%. The other 9 samples could not be detected. This implies that during the earliest warning point, most of the samples found to be a target-1 were warned for the first time.

Checkpoint	First Warning at Checkpoint	Cumulative Detection
25%	82	82%
50%	4	86%
75%	2	88%
100%	3	91%
Never detected	9	—

Table 5.3: Detection Rate and First-Warning Distribution

The average number of tokens left after receiving the first warning on the target-1 samples detected is 21.54 tokens and the median is 23. Minimum lead time = 0 tokens and the maximum lead time = 23 tokens. The values indicate that, in the many cases that are identified, the first warning was issued before a large percentage of the generation has been consumed.

Measure	Result
Target-1 samples	100
Detected by generation end	91
Detection rate	91%
Mean tokens remaining	21.54
Median tokens remaining	23
Minimum tokens remaining	0
Maximum tokens remaining	23

Table 5.4: Detection Lead-Time Results

5.8 RISK SCORE AND DETECTION EXAMPLES

The Logistic Regression detector returns a risk score which is either NORMAL or SUSPICIOUS depending on whether the score is above the fixed threshold of 0.5. To understand the behaviour of the detector during the actual inference process and the variation in the risks scores between the generation checkpoints, there will be some examples that will be retroactively discussed.

If the question asked is "What is the capital of France?," the answer generated contained repeat information about Paris, and there was some ambiguity about the location of Paris. The Final Risk was 0.3297, considered as NORMAL. An example of this via the prompt “Who was the first person to walk on the Moon?” resulted in GPT-2-medium providing the wrong answer, which was Custer, George Armstrong. The final risk score was, however, 0.1747 and NORMAL was set as the sample type.

The risk score was also demonstrated to be an active risk in a separate checkpoint experiment. A response, that had been generated on the question Moon-landing, had a risk score of 0.9228 for 25% of the responses, 0.3962 for 50% of the responses and 0.4901 for 75% of the responses (NORMAL and SUSPICIOUS respectively). This was raised slightly to 0.5021 at 100% which was deemed SUSPICIOUS (0.5020).

Input / Checkpoint	Tokens Observed	Risk Score	Classification
Capital of France — Final	31	0.3297	NORMAL
First person on Moon — Final	31	0.1747	NORMAL
Moon prompt — 25%	8	0.9228	SUSPICIOUS
Moon prompt — 50%	16	0.3962	NORMAL
Moon prompt — 75%	24	0.4901	NORMAL
Moon prompt — 100%	31	0.5021	SUSPICIOUS

Table 5.5: Example NeuroTrace Risk Scores

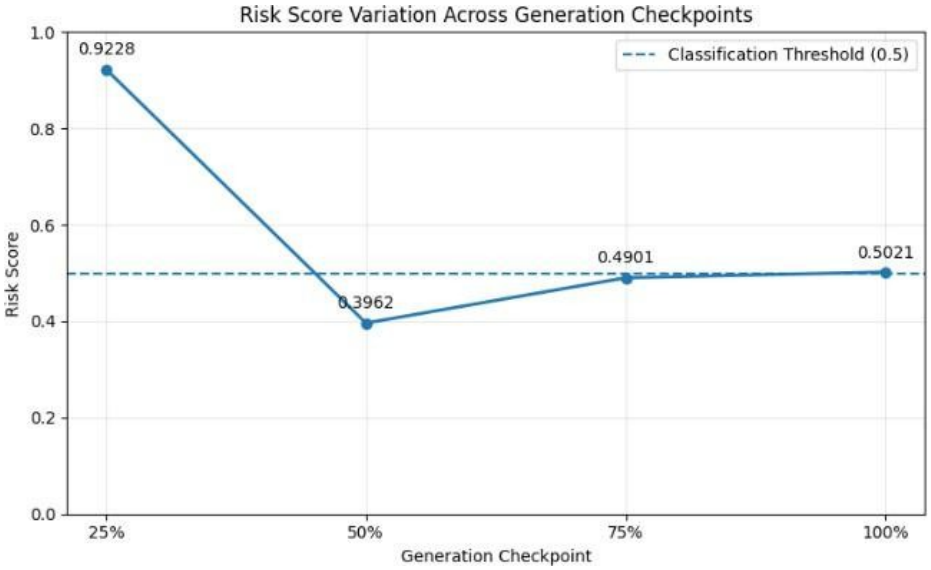


Figure 5.3: Risk Score Variation Across Generation Checkpoints

5.9 DISCUSSION

The Experimental results demonstrate that the hidden-state trajectory features can be useful for discriminating between the two generation conditions employed in NeuroTrace. In the ablation study, the trajectory-only configuration yielded the highest mean ROC-AUC (0.7700) compared to the uncertainty-only features (0.6030), indicating that internal representations of GPT-2-medium were a more informative signal, relative to token probability and entropy, in the controlled setting.

There is a restriction on this, however: The clearly incorrect answer of "I walked on the moon" was judged NORMAL (final risk score: 0.1747); The risk score ranged from 0.9228 at 25% to 0.3962 at 50% to 0.5021 at 100%. A high early risk score thus does not mean that there was actually an error, and a low score does not mean that there was not. In general, the results demonstrate feasibility of using temporal hidden-state trajectories as an experimental early-warning indicator. This is not a full hallucination-detection system, they show discrimination between the controlled conditions and this can be used as a basis for further research.

5.10 LIMITATIONS

There are some drawbacks to using NeuroTrace. The reported ROC-AUC, detection rate and lead-time measure discrimination between the two HaluEval-based conditions and not verified factual hallucination detection, which is carried out under controlled conditions. Only a small dataset (200 samples from 100 questions) is explored, and only layers 3-6 of a single frozen GPT-2-medium model are analysed ; the risk score could differ across checkpoints and even fail to capture specific factual errors as illustrated by the Moon-landing example. The system also models aggregated features instead of the full sequence with a simple Logistic Regression detector. The NeuroTrace is a research prototype, and needs to be tested with larger datasets, other models, with independently verified labels and sequential detectors.

CHAPTER 6
CONCLUSIONS AND SCOPE FOR FURTHER WORK

6.1 CONCLUSION

To understand if temporal internal changes of the hidden-state information that the frozen GPT-2-medium model maintains can serve as an early-warning signal for potentially anomalous generation, NeuroTrace was developed. The framework uses layers 3-6 to extract hidden states in process of generating tokens one-by-one and computes trajectory features like hidden-state transitions, cosine-similarity, relative state change, and hidden-state norms, in addition to token-level uncertainty features. These features were analyzed in controlled conditions with the help of the light-weight Logistic Regression detector and HaluEval was used for this purpose.

As shown in the experimental results, the trajectory-only features obtained a mean ROC-AUC of 0.7700, compared to the uncertainty-only features that obtained 0.6030. In the early-warning experiment, performance was found to be best at 25% (mean ROC-AUC 0.9095), indicating that there is some beneficial discriminative information that can be obtained before the generation. 91 % of the target-1 controlled-condition samples were detected at the fixed threshold of 0.5 and on the average 21.54 tokens were still not detected after the first warning was issued among the samples detected.

Meanwhile, the experiments show that the existing system shouldn't be regarded as a full-fledged hallucinator of facts. Not all target labels were necessarily accurate and individual examples showed that the detector might not detect the typical wrong answers, and may assign different risk scores at the checkpoints. Overall, the main take-away message of NeuroTrace is that, in contrast to being a reliable biomarker for factual hallucinatory detection, the temporal hidden-state trajectories are a promising signal to investigate as a biomarker for early monitoring.

6.2 KEY CONTRIBUTIONS

The main contribution of NeuroTrace is as follows:

1. Temporal Hidden-state Monitoring: we analyse the temporal evolution of the hidden-state representations of a "frozen" GPT-2-medium model when generating tokens sequentially.
2. Trajectory based feature extraction: The features are extracted from the layers (3-6), which represent the changes in the model's internal state, such as hidden-state norms, transition distances, cosine similarities and relative state changes.

3. Lightweight detection framework: The extracted feature set is used to analyse with a Logistic Regression detector with no fine tuning or modification of the language model.
4. The right-answer condition and the hallucinated-answer condition are created into a balanced dataset with 200 experimental samples, comprising of 100 questions each.
5. Configurations: The effects of different features are investigated by comparing the configurations "trajectory-only," "uncertainty-only," and "all-feature.
6. Early-warning analysis: Evaluation of the detection performance is done at 25%, 50%, 75% and 100% generation checkpoints instead of just after the full response.
7. Detection lead-time measurement: The number of tokens left after the first warning is sent will give a quantitative measurement of the lead-time of the detection signal.
8. Group-aware evaluation: StratifiedGroupKFold is used with the question/sample ID as the group, to avoid leakage of information between the paired experimental conditions.
9. Continuous risk score monitoring: Continuous risk score is generated by the system, which can be monitored during the whole generation process and can be identified as NORMAL or SUSPICIOUS by setting the fixed threshold.

The project also investigates cases of failure of the detector, and cases of varying risk scores, to give a more realistic idea of the feasibility and limitations of the use of temporal hidden-state monitoring.

6.3 CURRENT LIMITATIONS

NeuroTrace has a few limitations, the results of which must be interpreted with. This assessment is carried out on a controlled set of limited size (200 samples, 100 questions) which may not reflect performance on other types of prompts or actual LLM generation. Target conditions are not factually independent labels for hallucination, but rather self-generated from their own right in both HaluEval and hallucinated answers, respectively. The experiments also use only one frozen GPT-2-medium model, and only monitor layers 3-6, limiting the generalisation to other LLM architectures and sizes. Furthermore, the risk score may change from generation checkpoint to generation checkpoint and there are answers to questions that are obviously incorrect that are still scored as low-risk answers. The limitations proved that at present, NeuroTrace is a research prototype, which should be implemented with a larger amount of data, different language models, and labels from independent sources.

6.4 FUTURE WORK

There are a number of avenues for further development of NeuroTrace. This approach can be scaled to large and diverse controlled data sets with independently created factual labels,

in order to challenge the approach in more realistic hallucination detection scenarios. The framework can also be tested with different LLM architectures and size to determine whether temporal hidden-state patterns can be extended to other sizes of GPTs. Another approach would be to delve into various levels of granularity and sequential functionality instead of mainly aggregating statistics. Finally, the current checkpoint-based approach can be enhanced by taking temporal smoothing or sequential decision making approaches to decrease the variability in the risk score and generate more stable warning messages. Lastly, more advanced detection models and monitoring approaches for real-time systems can be explored, to determine whether hidden-state trajectories can provide a reliable early warning for deployment of LLM practically.

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APPENDIX A
IMPLEMENTATION DETAILS AND CONFIGURATION

The NeuroTrace framework was implemented using a frozen GPT-2-medium model for token generation and hidden-state extraction. Hidden-state representations from layers 3, 4, 5, and 6 were collected during token-by-token generation along with token probability and entropy values. These features were aggregated and provided to a Logistic Regression detector after standardization using StandardScaler. The trained detector was evaluated using StratifiedGroupKFold cross-validation and at 25%, 50%, 75%, and 100% generation checkpoints. The complete experimental configuration and software environment are listed in Tables 4.1 and 4.2 of Chapter 4.

APPENDIX B
FEATURE DEFINITIONS AND PARAMETERS

NeuroTrace uses hidden-state and uncertainty information extracted during token-by-token generation. The extracted features are converted into fixed-length response-level features using statistical aggregation. The main feature definitions and parameters used in the implementation are given below.

Table B.1: NeuroTrace Feature Definitions

Feature	Description
Hidden-State Norm	L2 magnitude of the hidden-state vector at each generation step
Hidden-State Transition	Euclidean distance between consecutive hidden-state vectors
Cosine Similarity	Directional similarity between consecutive hidden-state vectors
Relative State Change	Hidden-state transition normalized by the previous hidden-state norm
Token Probability	Probability assigned to the generated token
Entropy	Uncertainty of the model's output probability distribution
Mean	Average value of a feature across generated tokens
Standard Deviation	Variation of a feature across generated tokens
Maximum	Highest observed feature value
Minimum	Lowest observed feature value
Number of Tokens	Total number of generated tokens

The hidden-state trajectory features are extracted from layers 3, 4, 5, and 6 of GPT-2-medium. For each monitored layer, hidden-state norm, transition distance, cosine similarity, and relative state change are calculated. These features form the 40 trajectory features used in the trajectory-only detector.

Feature Group	Features Used	Number of Features
Trajectory-only	Hidden-state norm, transition, cosine similarity, relative state change	40
Uncertainty-only	Token probability, entropy	Uncertainty feature set
All features	Trajectory + uncertainty features	52 model features
Complete aggregated representation	All model features + number of tokens	53 columns

Table B.2: Feature Group Configuration

For each feature, the values observed across the generated tokens are summarized using mean, standard deviation, maximum, and minimum. These aggregated features are then standardized using StandardScaler before being provided to the Logistic Regression detector.

The early-warning analysis uses the same feature definitions at each generation checkpoint. Only the tokens available up to the selected checkpoint are used when calculating the corresponding aggregated features. This allows the detector to produce an intermediate risk score without requiring the complete generated response. The generation and detection parameters are listed in Table 4.1.

APPENDIX C
SAMPLE INPUT, OUTPUT AND ADDITIONAL RESULTS

This appendix presents representative examples of NeuroTrace inference results and checkpoint-wise risk scores. These examples demonstrate how the system generates a response, extracts internal features, and produces a risk score and classification.

C.1 Sample Input and Final Detection

The final risk scores for the two sample prompts are reported in Table 5.5. The first example produced a final risk score of 0.3297, which is below the classification threshold of 0.5 and was therefore classified as NORMAL. The second example demonstrates a limitation of the detector, as the generated response contained an incorrect answer but still received a low risk score of 0.1747.

C.2 Checkpoint-Wise Risk Score

The checkpoint-wise risk scores for the sample generation are reported in Table 5.5 and plotted in Figure 5.3. The checkpoint results show that the risk score can change during generation. In this example, the score was highest at the 25% checkpoint and subsequently decreased before increasing slightly at the final checkpoint. This demonstrates that the NeuroTrace risk score represents the detector's current output based on the available hidden-state trajectory and should not be interpreted as a direct measure of factual correctness.

C.3 Additional Early-Warning Results

The mean ROC-AUC values and cumulative detection rates at each checkpoint are given in Tables 5.2 and 5.3. The results show that the highest mean ROC-AUC was obtained at the 25% checkpoint, while the cumulative detection rate increased as generation progressed. Among the target-1 controlled-condition samples detected by the end of generation, the mean number of tokens remaining after the first warning was 21.54, with a median of 23 tokens.